

# Harsh Tiwari

Senior Software Engineer — Bengaluru, India

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## EXPERIENCE

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### WellnessLiving — Senior Software Engineer

Bengaluru, India — April 2025 - Present

- Engineered a **multi-tenant Voice AI backend** for production agents with **RAG**, structured **function calling**, tool schemas, and provider integrations, enabling tenant-specific workflows with reliable LLM-driven actions across live customer conversations.
- Built a production-ready **Chinese voice-to-voice agent** using **LiveKit** as the real-time media binding layer, self-deploying **Qwen3-ASR** for STT and **Qwen3-TTS** for speech synthesis with **CUDA/GPU inference**, while routing reasoning and tool orchestration through **Gemini**.
- Re-architected **Cerebrum**, the LLM microservice behind telephony agents, to deliver **sub-second responses** by streaming model output, parallelizing retrieval and tool preparation, trimming prompt-construction overhead, and removing synchronous bottlenecks from the inference path.
- Improved agent execution reliability with service-level guardrails for **tool execution**, structured tracing, regression checks, retry-safe workflows, and read-only SQL-agent constraints, reducing failure risk across RAG-backed analytics and live voice-agent workflows.
- Scaled a **WebSocket platform** with stateless services, **OAuth 2.0** authentication, and **rate limiting**, improving connection stability for real-time chat and agent workflows across tenant-facing product surfaces.
- Deployed **production-grade latency instrumentation** across **5 stages** of the voice path: **STT**, request assembly, **LLM inference**, tool execution, and **TTS**, using **structured tracing** and service-level timing hooks, streaming telemetry through **Kafka** into **Redshift** for tail-latency attribution and **sub-second response optimizations**.

### Kusho — Full Stack Developer

Bengaluru, India — August 2023 - April 2025

- Built **Redis and RQ-based worker pipelines** for scalable **LLM task execution**, prompt processing, and **asynchronous job orchestration** across large API payloads, improving throughput under heavy background workloads.
- Designed a unified **service-definition DSL** for **4 API specification formats**: **Postman**, **OpenAPI/Swagger**, **GraphQL**, and **Protocol Buffers**, then integrated **LLM APIs** to auto-generate test cases and reduce test authoring time.
- Built an **API testing and execution platform** with **50+ workflow node types** for end-to-end orchestration, improving large-flow execution performance and shortening test runtime for complex workflows.
- Developed a **Chrome extension** that records user interactions and generates **Playwright test scripts**, implemented robust selectors with fallback strategies for UI changes, and delivered **95% stability**, **3x test coverage**, and **70% less manual QA**.
- Built a **distributed test execution engine** that modeled large API collections as dependency-aware **workflow graphs**, executing nodes via **Redis based worker queues (RQ)** with dependency-aware scheduling, parallel fan-out, and isolated execution pipelines to support high-concurrency workflow simulation across **1000+ businesses**.
- Built a **compiler** that normalized **4 specification formats** including **Postman**, **OpenAPI**, **GraphQL**, and **Protobuf** into a common executable graph IR, enabling a **shared runtime** for test generation, validation, and large-scale API workflow execution.

### Castler — Full Stack Developer

Delhi, India — May 2021 - August 2023

- Built an **event-driven payment platform** for **UPI/IMPS** using message queues, idempotency controls, and circuit breakers, processing **INR 50Cr+ monthly** across **2M+ transactions** with **sub-200ms latency** while reducing infrastructure cost by **35%**.
- Architected a **fault-tolerant payee validation pipeline** that absorbed **1000+ RPM peak demand** against a heavily rate-limited upstream bank API, using asynchronous queueing, deduplication, load-regulation controls, and stateful retries to avoid user-facing failures during rush-hour traffic.
- Led a **zero-downtime migration** of **200GB** of financial data from **MongoDB** to **MySQL** using blue-green deployment and dual-write architecture, then built **Airflow reconciliation pipelines** that improved data consistency, delivered **40% faster queries**, and reduced infrastructure cost by **25%**.
- Strengthened **OAuth 2.0/PKCE authentication** and automated security scanning with **SonarQube**, **Snyk**, and **OWASP ZAP** to improve application security, release hygiene, and compliance readiness.
- Built **automated state-reconciliation and recovery workflows** across **3 sources of truth**: gateway responses, asynchronous bank callbacks, and internal ledger state to detect mismatched transaction states, recover failed payment flows, and preserve **financial correctness at scale**.

## SKILLS

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**Languages:** Python, TypeScript, Go, SQL

**Backend & APIs:** Node.js, NestJS, REST APIs, GraphQL, WebSockets, OAuth 2.0, Rate Limiting, Microservices, Event-Driven Architecture, Distributed Systems

**Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, Kafka, Redshift, Airbyte, AWS, Docker, Kubernetes, CI/CD

**AI/ML:** LLM APIs, Gemini, Agentic Workflows, Function Calling, Tool Execution, RAG, Vector Databases, Qwen3-ASR/TTS, LiveKit, CUDA/GPU Inference, Prompt Engineering, AI Evaluation

## EDUCATION

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**Amity University** — Bachelor of Technology in Computer Science

Lucknow, India — Aug. 2017 – May 2021